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Research Insights

Synthesis and photovoltaic properties of the efficient polymer donor based on benzo[1,2-*b*:4,5-*b*']difuran unit

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ABSTRACT

Benzo[1,2-*b*:4,5-*b*']dithiophene (BDT) has been used as the electron-donating block in the majority of polymer donors which generally achieve high-performance organic solar cells (OSCs). Benzo[1,2-*b*:4,5-*b*']difuran (BDF), with a similar structure to BDT, has attracted increasing attention because its raw materials are renewable. To diversify building blocks for high-performance donor polymers, we synthesized a new polymer donor, namely PFD, based on the BDF and benzo[1,2-*c*:4,5-*c*']dithiophene-4,8-dione (BDD) units as the backbone with 1-(2-butyloctyl)-chlorobenzene as the side-chain. The PFD possesses a deeper highest occupied molecular orbital (HOMO) energy level of −5.54 eV, which helps to generate a high open-circuit voltage (V_{oc}) in OSCs. When Y6 was chosen as an acceptor, a fair V_{oc} of 0.867 V and a decent PCE of 16.95 % were achieved in the binary OSCs. With the addition of a small molecule acceptor BTMe-4Cl as the third component, the PCE of the PFD:Y6:BTMe-4Cl ternary device has improved to 17.58 %. The results indicate that BDF-based polymers exhibit the potential for fabricating efficient OSCs.

1. Introduction

Organic solar cells (OSCs) have attracted much attention due to their advantages of being lightweight, flexible, and capable of large-area printing, which can be used in the solar power-generating window for building or vehicle integration, agricultural photovoltaic greenhouses and powering smart wearables [1–7]. Thanks to the rapid development of non-fullerene acceptors (NFAs) [8], and continuous innovation of interface materials and device processing methods [9,10], the power conversion efficiencies (PCEs) of the state-of-the-art OSCs have reached 20 % [11–14]. The active layer of OSCs typically consists of donor and acceptor, so the high-performance polymer donor is also one of the main factors determining the power conversion efficiency (PCE) of OSCs. Therefore, it is necessary to design and synthesize the polymer donors that possess matched physical properties with NFAs, such as deeper highest occupied molecular orbital (HOMO) to maximize higher open-circuit voltage (V_{oc}) of OSCs, and good miscibility with NFAs to form proper phase separation [15–17].

At present, one of the most widely used donor materials is benzo[1,2-*b*:4,5-*b*']dithiophene (BDT)-based polymers [18–20]. In 2012, benzo[1,2-*b*:4,5-*b*']difuran (BDF), a new building block with a similar structure to BDT was first reported for OSCs by Huo et al., which exhibited a PCE of 5 % [21]. Since then, BDF has attracted increasing attention. To comprehensively evaluate the properties of the BDF copolymer, Huo et al. synthesized PBDF-T1 and its BDT counterpart (PBDT-T1). Benefiting from the greater electronegativity of oxygen atom than that of sulfur atom, PBDF-T1 exhibits a deeper HOMO energy level, resulting in an increasing V_{oc} and a higher PCE of 9.43 % for its corresponding device than those of PBDT-T1 OSCs (PCE = 8.12 %) [22]. The oxygen atom possesses a smaller radius than the sulfur atom, improving the planarity of BDF-based polymer backbones compared with that of the BDT-based ones and thus compressing intermolecular stacking, which enhances film crystallinity and carrier mobility in the devices [23–28]. On the other hand, the raw material for BDT units is derived from petrochemical production [29–31]. Whereas, BDF blocks may be from an abundant renewable resource, furfural, collected from hemicellulose-rich agricultural wastes [32–34]. Such an environmentally friendly upstream material may be more suitable for large-scale production [35–38]. These benefits highlight the importance of developing high-performance BDF-based polymer donor materials.

We have previously reported a BDF-based polymer donor PBF1-C, which is composed of BDF and benzo[1,2-*c*:4,5-*c*']dithiophene-4,8-dione connected by thiophene π-bridge [39]. Chlorination is a promising strategy that could effectively tune the energy levels and well control the

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morphologies for OSCs [40]. On this basis, we synthesized a new polymer donor PFD by introducing chlorine atoms on the phenylic side chain. The HOMO energy level of the PFD is deeper at −5.54 eV than that of the PBF1-C (−5.43 eV). After blending with Y6, the PCE of the binary device reached 16.95 % with an obviously enhanced V_{oc} from 0.74 V to 0.868 V, which is comparable to the PCE (17.0 %) of the binary device based on PM7, a structurally similar BDT analogue [41]. In addition, the ternary strategy has advantages in terms of the broadening of the absorption spectral range and optimization of the active layer morphology [42–44]. We have introduced a third component acceptor BTMe-4Cl, which shows a significant blue-shifted absorption region compared to Y6, a cascade energy levels with PFD and Y6, and also has a suitable phase separation morphology in the PFD:Y6:BTMe-4Cl system, improved the V_{oc} and short-circuit current J_{sc} and led to a PCE of 17.58 % for the ternary devices. This work demonstrates that BDF-based polymer donors exhibit a bright future in the development of efficient OSCs.

2. Methods

2.1. Materials and synthesis

All materials and solvents were purchased from Tianjin Kemiou Chemical Reagent, Alfa Aesar and used without further purifications. The synthesis routes of the polymer donor (PFD) and acceptor BTMe-4Cl were shown in Scheme 1 and Scheme 2. The compound **1, 6,** and **7** were purchased from Alfa Aesar, SunTech Inc, respectively. The Y6 was purchased from Solarmer Inc. The compounds **2** and **3** were synthesized according to relevant literature [45], and compound **8** was synthesized according to relevant literature [46].

2.1.1. Synthesis of compounds *4*

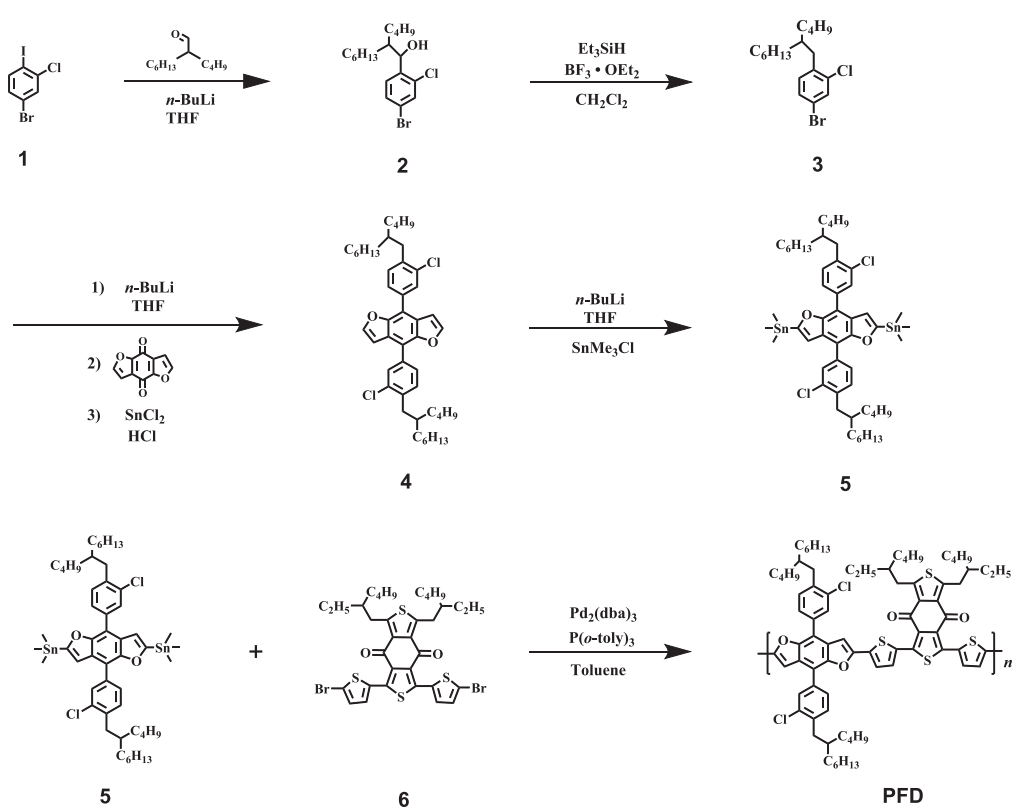
Compound **3** (3.62 g, 10.06 mmol) was dissolved in anhydrous tetrahydrofuran (25 mL) and stirred at −78 °C under Ar atmosphere. After *n*-BuLi (4.1 mL, 10.25 mmol, 1.02 eq, 2.5 M in *n*-hexane) was added dropwise, the solution was kept at the same temperature for 2.5 h. After that benzo[1,2-*b*:4,5-*b*']difuran-4,8-dione (0.95 g, 5.03 mmol, 0.5 eq) was added, the mixture was stirred at room temperature for 6 h. Then $SnCl_2$ (8.40 g, 44.28 mmol, 4.4 eq) in hydrochloric acid solution (10 %, 16.8 mL) was added and the mixture was stirred overnight. The product was extracted with dichloromethane and then dried with anhydrous Na_2SO_4. After filtering and concentrating, the crude product was further purified through column chromatography (eluent: petroleum ether) to obtain a colorless liquid (0.84 g, yield: 23.5 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ(ppm): 7.86 (s, 2H), 7.73 (d, J = 2.1 Hz, 2H), 7.67 (d, J = 9.1 Hz, 2H), 7.36 (d, J = 7.9 Hz, 2H), 7.04 (d, J = 2.1 Hz, 2H), 2.75 (d, J = 7.1 Hz, 4H), 1.89–1.78 (m, 2H), 1.38–1.27 (m, 32H), 0.93–0.87 (m, 12H). MALDI-TOF MS (*m/z*): calcd. for $C_{46}H_{60}Cl_2O_2$: 715.884; Found: 715.673.

2.1.2. Synthesis of compound *5*

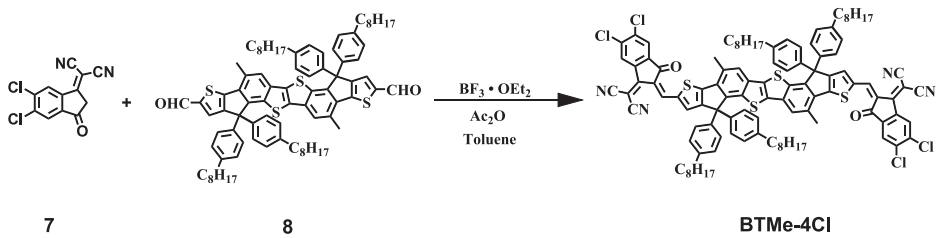
Compound **4** (0.52 g, 0.73 mmol) was dissolved in anhydrous tetrahydrofuran (12 mL) and stirred under Ar atmosphere. After *n*-BuLi (0.96 mL, 2.4 mmol, 3.28 eq, 2.5 M in *n*-hexane) was added dropwise, the solution was stirred for 2 h at 0°C. Trimethyltin chloride solution (1.0 M in *n*-hexane, 3 mL) was added to the reactant at −78 °C, afterward, the mixture was stirred overnight at room temperature. The product was extracted with petroleum, and then dried with anhydrous Na_2SO_4. After filtering and concentrating, the crude product was further purified by recrystallization from ethanol/methanol. A light yellow solid was obtained after drying in vacuo (0.55 g, yield: 73.1 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ (ppm): 7.94: (s, 2H), 7.73 (d, J = 7.9 Hz, 2H), 7.36 (d, J = 7.9 Hz, 2H), 7.19 (s, 2H), 2.75 (d, J = 7.0 Hz, 4H), 1.91–1.79 (m, 2H), 1.38–1.26 (m, 32H), 0.94–0.87 (m, 12H), 0.42 (s, 18H). MALDI-TOF MS (*m/z*): calcd. for $C_{52}H_{76}Cl_2O_2Sn_2$: 1041.498; Found: 1041.645.

2.1.3. Synthesis of polymer *PFD*

The compound **5** (34.7 mg, 0.03 mmol), compound **6** (26.2 mg, 0.03



Scheme 1. Synthesis route of polymer PFD.



Scheme 2. The synthetic route of acceptor BTMe-4Cl.

mmol), $Pd_2(dba)_3$ (1.2 mg, 0.0013 mmol), and P(*o*-tol)$_3$ (3.3 mg, 0.0107 mmol) were dissolved in anhydrous toluene (1.7 mL), followed by purging with argon for 30 min in liquid nitrogen bath. Under Ar atmosphere, the system was heated to 110 °C at 24 h. The solution was added dropwise into ethanol (30 mL) and the precipitate was collected by filtration. Subsequently, the crude polymer was purified through Soxhlet extraction using petroleum ether, dichloromethane, and chloroform in turn. The chloroform fraction was concentrated, and a dark blue solid was obtained after drying in vacuo (32.5 mg, yield: 53.4 %). Molecular weight by GPC (chloroform, room temperature): M_n: 294.2 kDa, PDI = 2.13. Molecular weight by GPC (1,2,4-trichlorobenzene, 150 °C): M_n: 30.8 kDa, PDI = 4.73.

2.1.4. Synthesis of small molecule acceptor ***BTMe-4Cl***

To a two-necked round-bottom flask were added compound **7** (26.1 mg, 0.0988 mmol), compound **8** (50.0 mg, 0.0395 mmol), and 5 mL toluene. Under argon, 0.05 mL BF_3•OEt_2 and 0.08 mL Ac_2O were added, and the mixture was stirred overnight. Afterward, the solution was added dropwise into methanol and the crude product was collected by filtration. The obtained filter residue was purified by column chromatography (petroleum ether/dichloromethane, v/v = 2:1) to obtain a blue-black solid (33.0 mg, yield: 47.5 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ(ppm): 8.90 (s, 2H), 8.77 (s, 2H), 7.97 (s, 2H), 7.82 (s, 2H), 7.72 (s, 2H), 7.13 (d, J = 8.3 Hz, 8H), 7.08 (d, J = 8.3 Hz, 8H), 2.88 (s, 6H), 2.55 (t, J = 7.8 Hz, 8H), 1.56–1.51 (m, 4H), 1.37–1.14 (m, 42H), 0.89–0.81 (m, 12H).

MALDI-TOF MS (*m/z*): calcd. for $C_{108}H_{100}Cl_4N_4O_2S_4$: 1756.050; Found: 1756.715.

2.2. Characterization

The ^{1}H NMR of all compounds were acquired on a Bruker AVANCE 400 MHz spectrometer, using deuterated chloroform ($CDCl_3$) as solvent. Mass spectrometry (MS) was measured on a Bruker Aupoflex III MALDI-TOF analyzer. The molecular weights of the polymers were determined by gel permeation chromatography (GPC) using trichloromethane as eluent at room temperature and 1,2,4-trichlorobenzene as the eluent at 150 °C, respectively. The thermogravimetric analysis (TGA) was carried out on a TA Instruments TGA-50 thermogravimetric analyzer with nitrogen at a scanning rate of 15°C min^{-1}. Photophysical properties were performed on a Perkin-Elmer Cary 60 UV–Vis-NIR spectrophotometer. Electrochemical properties were determined using cyclic voltammetry (CV) in an electrochemical workstation (Shanghai Chenhua Electrochemical Workstation CHI660D). And the three-electrode cell consisted of ITO glass coated with a donor film as the working electrode, Ag/AgCl as the reference electrode, platinum wire as the auxiliary electrode, and ferrocene as the internal standard. The CV was measured in acetonitrile solution containing Bu_4NPF_6 (0.1 M) as the electrolyte, with the sweep rate at 100 mV s^{-1}. Photoluminescence (PL) was performed on the F-4600 fluorescence spectrophotometer. The contact angle was tested by the POWEREACH JC200D1 contact angle measuring instrument. Atomic force microscopy (AFM) images were acquired on the MUITIMODEPI-COFOREE atomic force microscope in a tapping mode.

2.3. Device fabrication and measurements

All the devices were fabricated by using the conventional structure ITO/PEDOT:PSS/active layer/PNDIT-F3N/Ag. Acetone, water, Acetone, and isopropanol were used to clean the ITO glass in turn. The ITO substrates were exposed to UV-ozone for 15 min. After spin-coating the PEDOT:PSS solution onto the them, ITO glass substrates were annealed for 15 min at 150°C. The PFD:Y6, PFD:BTMe-4Cl and PFD:Y6:BTMe-4Cl solution in chloroform (15 mg mL^{-1} in total) with 0.5 vol% 1-chloronaphthalene (1-CN) as the additive were spin-coated on the substrates, respectively, and then annealing at 100 °C for 10 min. Subsequently, a thin layer of PNDIT-F3N (1mg m L^{-1} in methanol with 0.25 vol% acetic acid) was spin-coated onto the active layer at a rate of 4000 rpm for 35 s. Eventually, Ag (100 nm), used as an electrode, was thermally evaporated at a pressure of 2.2 × 10^{-4} Pa. The effective area of devices is 0.038 cm^2. The current–voltage (*J-V*) characteristics of the solar cell were measured with a Keithley 2602 source meter and an Oriel Sol3A solar simulator equipped with an AM 1.5G solar spectral filter with an intensity of 100 mW cm^{-2} as a light source. The external quantum efficiency (EQE) spectra are characterized using the Zolix Solar Cell Scan 100 QE/IPCE measurement system. The space-charge-limited current (SCLC) method was used to characterize the hole mobility (μ_h) and electron mobility (μ_e) of the blend films. The device structures for hole mobility are ITO/PEDOT:PSS/active layer/MoO_3/Ag, and for electron mobility are ITO/ZnO/active layer/PNDIT-F3N/Ag. The *J-V* characteristics of hole-only and electron-only devices fitted by the Mott-Gurney equation are shown below:

$$J = \frac{9}{8}\varepsilon_r\varepsilon_0\mu\frac{V^2}{L^3}$$

where J is the current density, L is the thickness of the active layer, ε_0 is the permittivity of free space, ε_r is the relative permittivity, and V is the applied voltage.

3. Results and discussions

3.1. Synthesis and characterization

The molecular structures of the polymer donor PFD and the small molecule acceptor BTMe-4Cl are shown in Fig. 2a, and the specific synthetic methods are shown in Scheme 1 and Scheme 2. The intermediates **3** and **8** were synthesized according to the reported literature [45,46]. Intermediate **3** was reacted with benzo[1,2-*b*:4,5-*b*'] difuran-4,8-dione to obtain compound **4**. Compound **4** was deprotonated with *n*-BuLi and then quenched by trimethyltin chloride to get compound **5**. The target polymer PFD was easily obtained by palladium-catalyzed Stille coupling polymerization between monomers **5** and **6**. Compounds **7** and **8** in toluene were reacted with two reagents BF_3•OEt_2 and Ac_2O under Ar atmosphere to give the small molecule acceptor BTMe-4Cl. The chemical structures of all the above compounds were confirmed by ^{1}H NMR and MALDI-TOF MS as shown in Figs. S1–S6 (Supporting Information). PFD and BTMe-4Cl are well soluble in common organic solvents such as chloroform and chlorobenzene. The number-averaged molecular weight (M_n) and the polydispersity index

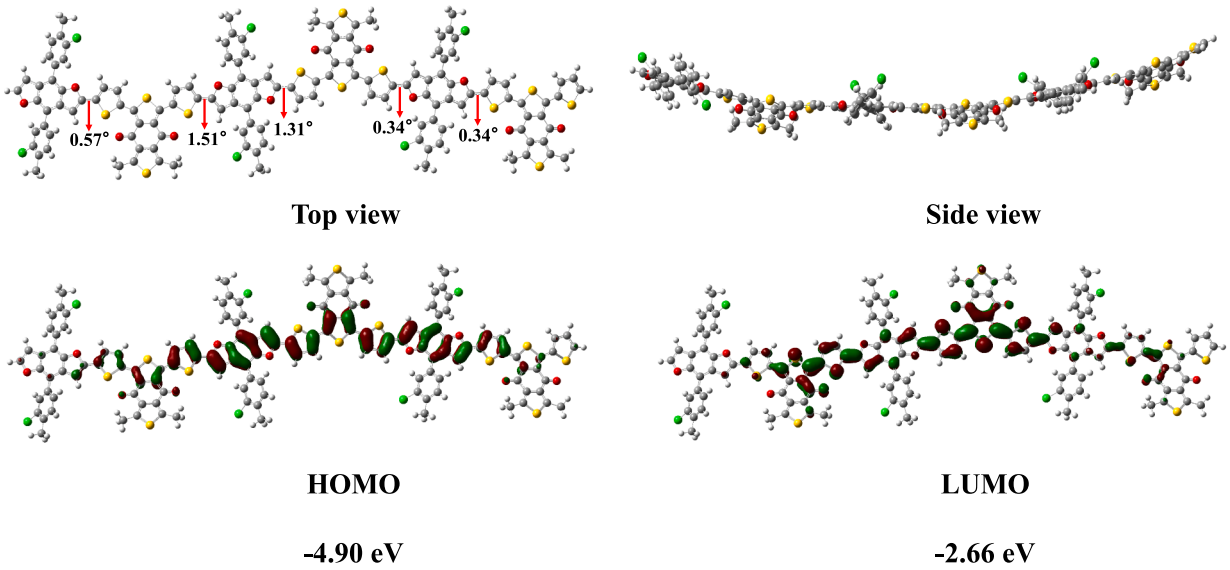


Fig. 1. Simulated molecular geometries and frontier molecular orbital distribution of PFD.

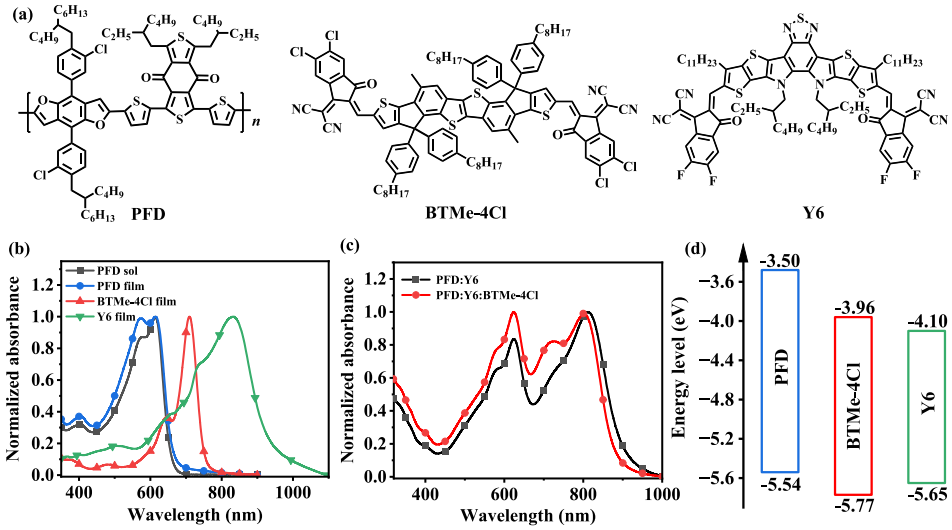


Fig. 2. (a) Molecular structures of PFD, BTMe-4Cl, and Y6. (b) Normalized UV–vis absorption spectra of PFD, BTMe-4Cl, and Y6, and (c) PFD:Y6 and PFD:Y6:BTMe-4Cl blend films. (d) Frontier molecular orbital energy levels of PFD, BTMe-4Cl, and Y6.

(PDI) of PFD are 294.2 kDa and 2.13 when trichloromethane was used as the eluent at room temperature (Fig. S7). On the contrary, a lower M_n of 30.8 kDa and a PDI of 4.73 can be obtained under 1,2,4-trichlorobenzene as the eluent at 150 °C, indicating that PFD can aggregate strongly in chloroform solution (Fig. S8). As shown in Fig. S9, the thermal decomposition temperature (T_d, 5 % weight loss) of the PFD was 413°C under a nitrogen atmosphere. The high decomposition temperature indicates the outstanding processing stability of the PFD.

3.2. Theoretical calculations

The optimized molecular geometry of PFD with simplified side chains was calculated by density functional theory (DFT) at the B3LYP/6-31G (d) level. As shown in Fig. 1, the dihedral angles between the BDF units and the thiophene bridges do not exceed 1.51°, indicating that the molecular backbone has good planarity, which is favorable for molecular packing. The electron cloud in the HOMO is delocalized throughout the backbone, while the LUMO electron cloud is concentrated on the acceptor units, implying that such structure is conducive to carrier transport. Additionally, the HOMO/LUMO energy levels of PFD were calculated to be −4.90/−2.66 eV.

3.3. Photophysical and electrochemical properties

The ability to trap sunlight was characterized by normalized ultraviolet–visible (UV–vis) absorption spectra as shown in Fig. 2b-2c, and the corresponding data are given in Table 1. The PFD possesses a wide absorption range between 350 and 650 nm. Compared with its

Table 1
Photophysical and electrochemical properties of donor and acceptor materials.

Materials	$\lambda_{sol\ max}$ (nm)	$\lambda_{film\ max}$ (nm)	λ_{edge} (nm)	*E*opt g (eV)	LUMO (eV)	HOMO (eV)	E_g (eV)
PFD	614	614	653	1.90	−3.50	−5.54	2.04
BTMe-4Cl	–	708	765	1.62	−3.96	−5.77	1.81
Y6	–	834	932	1.33	−4.10	−5.65	1.55

absorption in chloroform solution, the absorption in film hardly changed, showing that PFD polymer has strong molecular pre-aggregation in solution [47]. The absorption of the PFD:Y6 blend film is weak around 650–750 nm due to the absorption of Y6 mainly distributed in the 700–900 nm range. Therefore, we selected a third component (BTMe-4Cl) with absorption exactly in the mid-wavelength region (550–800 nm) to supplement the absorption of the PFD:Y6 binary film, which improves the photon trapping ability of the device and thus conduces to the enhancement of J_{sc}.

The electrochemical properties of the polymer donor PFD, small molecule acceptors BTMe-4Cl and Y6 were tested by CV measurement. The CV curves are shown in Fig. S10. The energy levels of PFD, BTMe-4Cl, and Y6 are shown schematically in Fig. 2d and the specific data are summarized in Table 1. The HOMO and lowest unoccupied molecular orbital (LUMO) energy levels of the PFD are −5.54 eV and −3.50 eV, which match well with those (−5.65 eV and −4.10 eV) of the typical narrow bandgap acceptor Y6. The HOMO and LUMO energy levels of the third component, BTMe-4Cl, are −5.77 eV and −3.96 eV. In addition, the cascade energy level arrangement between PFD, BTMe-4Cl and Y6, facilitates charge transfer, resulting in a higher V_{oc} in the OSC [48,49].

3.4. Photovoltaic properties

The photovoltaic performance of all devices was characterized under the irradiation at AM 1.5G (100 mW cm^{-2}) and based on ITO/PEDOT:PSS/active layer/PNDIT-F3N/Ag structure. The binary devices based on PFD:Y6 were systemically optimized by varying the D/A ratio (Table S1, Supporting Information), annealing temperature, and additives (Table S2). The *J-V* curves for the best conditions are shown in Fig. 3a and the corresponding photovoltaic parameters are listed in Table 2. The optimal devices exhibited the maximum PCE of 16.95 % with a high V_{oc} of 0.868 V, a J_{sc} of 25.59 mA cm^{-2}, and an FF of 76.58 %. Such efficiency can bear comparison with that (PCE of 17.0 %) of the PM7:Y6 binary device [41]. To further enhance the PCE, we introduced the third component acceptor (BTMe-4Cl) due to its appropriate absorption and energy level. When the ratio of PFD:Y6:BTMe-4Cl was 1:1.2:0.2, the optimum value of PCE was 17.58 % thanks to the increased V_{oc} and J_{sc} (Table S3).

To provide feedback on the optical response of each device, the EQE measurement was carried out, and Fig. 3b shows the EQE and integral current curves of the binary and ternary OSCs. Both binary and ternary devices reveal a broad and strong photon response in the wavelength range of 300–900 nm. Compared with the $J_{sc,cal}$ of 24.57 mA cm^{-2} for the PFD:Y6 device, the addition of BTMe-4Cl enhances the photoresponse of the ternary OSC in the range of 420–820 nm, achieving a $J_{sc,cal}$ of 25.49 mA cm^{-2}. The $J_{sc,cal}$ value from the EQE spectra agrees with the measured J_{sc} within 5 % mismatch, confirming the credibility of the J_{sc} values obtained from the *J-V* curves [50].

Device stability is also an important consideration for the practical application of OSCs. We investigated the photo-stability of PFD:Y6 and PFD:Y6:BTMe-4Cl encapsulated devices under white light illumination of 1 sun intensity in air. As shown in Fig. S11, both the time of PCE reduction to 80 % (T_{80}) was 140 h. Additionally, we measured the

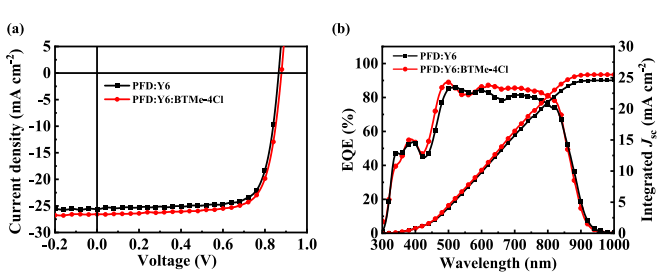


Fig. 3. (a) *J–V* curves and (b) EQE spectra of based on PFD:Y6 and PFD:Y6:BTMe-4Cl OSCs.

Table 2
Photovoltaic performance of optimized binary and ternary OSCs.

PFD:Y6:BTMe-4Cl	V_{oc}[c] (V)	J_{sc}[c] (mA cm^{-2})	FF[c] (%)	PCE[c] (%)
1:1.2:0[a,b]	0.868 (0.864 ± 0.006)	25.59 (25.60 ± 0.28)	76.58 (76.10 ± 1.05)	16.95 (16.83 ± 0.09)
1:1.2:0.2[a,b]	0.878 (0.878 ± 0.004)	26.55 (26.32 ± 0.24)	75.39 (75.17 ± 0.47)	17.58 (17.40 ± 0.21)

[a] Annealing for 10 min at 100°C.
[b] 0.5% 1-CN additive.
[c] The maximum values and the average values with standard deviations were obtained from ten devices.

thermal and long-term stability of the PFD:Y6 and PFD:Y6:BTMe-4Cl unpackaged devices in a nitrogen-filled glove box. At a temperature of 65 °C, the T_{80} lifetime for binary and ternary devices was 850 h and 1000h, respectively, indicating that the incorporation of BTMe-4Cl can improve the thermal stability of the ternary OSCs. The corresponding devices can maintain over 95 % of the initial PCE after 2000 h at room temperature without light, which means that both binary and ternary PFD-based devices exhibit excellent long-term storage stability.

Charge transport properties of the OSCs are an essential factor that affects the performance of photovoltaics, in which the SCLC measurement was used to characterize the hole and electron mobility in the blend films. From Fig. 4a–4b and Table S4, the μ_h/μ_e of the PFD:Y6 and PFD:Y6:BTMe-4Cl blend films were calculated to be $4.01 \times 10^{-4}/3.70 \times 10^{-4}$ and $4.77 \times 10^{-4}/4.63 \times 10^{-4}$ cm^2 V^{-1} s^{-1}, with μ_h/μ_e ratios of 1.08 and 1.03, respectively. The higher carrier mobilities and more balanced μ_h/μ_e ratios could contribute to an enhanced J_{sc} [51,52].

To further understand the exciton dissociation and charge collection efficiency, we analyzed the photocurrent density-effective voltage (J_{ph}-V_{eff}) curves (Fig. S12). The J_{ph} at short circuit and maximum power output were defined as *J*a ph and *J*b ph, respectively, and the exciton dissociation (P_{diss}) and charge collection (P_{coll}) efficiencies were calculated from the formulas $P_{diss} = Ja\ ph/J_{sat}$ and $P_{coll} = Jb\ ph/J_{sat}$ (J_{sat} is the saturation current density). The P_{diss}/P_{coll} of the devices based on PFD:Y6 and PFD:Y6:BTMe-4Cl are 97.41 %/90.21 % and 98.01 %/91.35 %, respectively. The efficient exciton dissociation and charge collection are the keys to the high photovoltaic performance of the devices [53].

The dependence of J_{sc} and V_{oc} on light intensity was studied to investigate charge recombination in the active layer. The equation, $V_{oc} \propto n\ (k_BT/q)\ \ln P_{light}$, is used to reflect the mechanism of charge recombination. The slopes of PFD:Y6 and PFD:Y6:BTMe-4Cl are 1.110 and 1.056 k_BT/q, as displayed in Fig. 4c. The slopes are much closer to 1, indicating that the main recombination mechanism in both active layers is bimolecular recombination and trap-assisted recombination is suppressed. The J_{sc} and P_{light} satisfy the equation: $J_{sc} \propto P_{light}^{\alpha}$, the value of α reflects the strength of bimolecular recombination. When the α is close to 1, the degree of bimolecular recombination is negligible. The α values of 0.998 and 0.999 for PFD:Y6 and PFD:Y6:BTMe-4Cl, respectively, imply that the recombination rarely occurs in the two OSCs [54,55].

The exciton dissociation and charge transfer of the donor and acceptor can be verified by PL spectroscopy. The PL results of the pure polymer donor film and all the blended films based on PFD:acceptors are shown in Fig. S13, where the excitation wavelength is 612 nm. For the pure PFD film, a PL peak can be found at 699 nm. When PFD was blended with Y6 and BTMe-4Cl in the ratios of 1:1.2:0 and 1:1.2:0.2, the emission peaks of both blended films were significantly reduced. In detail, the fluorescence quenching efficiencies of the binary and ternary blend films are 98.41 % and 98.85 %, respectively. The higher efficiency means better exciton dissociation and charge transfer from donor to acceptor, which helps to obtain a higher J_{sc} [56,57].

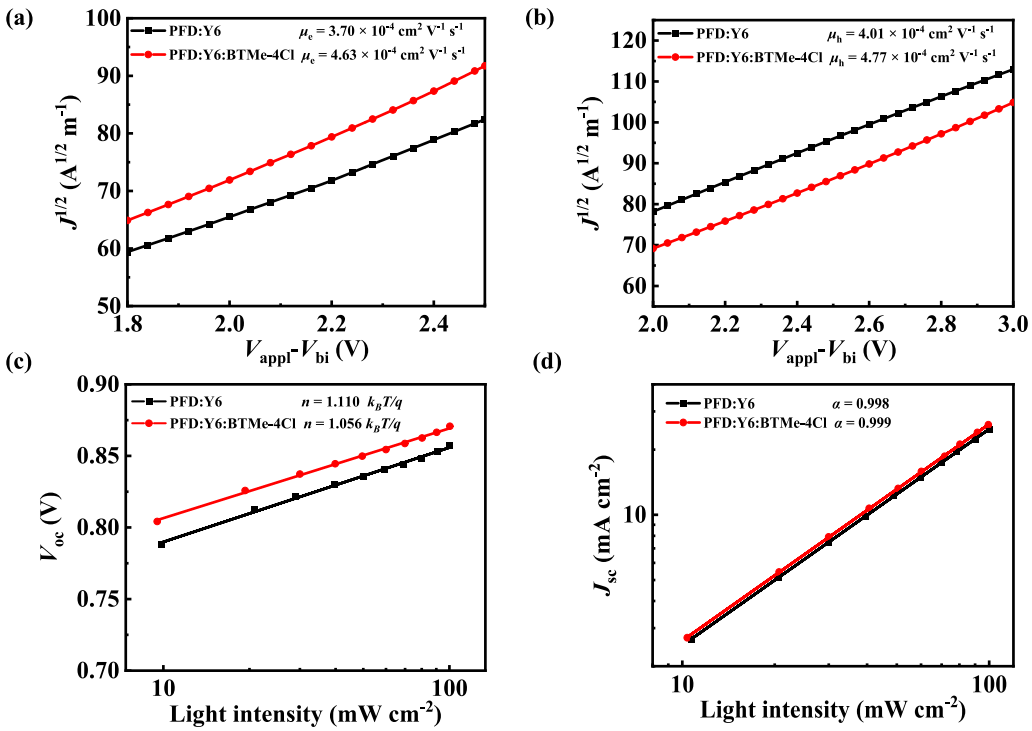


Fig. 4. $J^{1/2}$–V curves of (a) electron-only and (b) hole-only based on the binary and ternary devices. Dependence of (c) V_{oc} and (d) J_{sc} on the light intensity of the binary and ternary devices.

3.5. Film morphology

The proper compatibility between materials is the premise of high-quality film morphology. We performed the contact angles of water and ethylene glycol on the PFD, Y6, and BTMe-4Cl pure films, and the surface energy values and the Flory-Huggins interaction parameter (χ) were calculated to evaluate the factors among them, according to the Owens-Wendt-Kaelble's model and the formula: $\chi = k\,(\sqrt{\gamma_D} - \sqrt{\gamma_A})^2$. Fig. S14 and Table S5 show the relevant data. The χ of PFD:Y6 is 0.0470 k, indicating that there was good compatibility between them, which avoids a serious aggregation. With respect to the third component BTMe-4Cl, a larger χ of 0.0667 k can be obtained for PFD:BTMe-4Cl, while the χ of Y6:BTMe-4Cl is as small as 0.0017 k. The result suggests that Y6 and BTMe-4Cl can form a well-mixed acceptor phase in the ternary blends, and BTMe-4Cl can conduct an improved phase separation by repulsing the donor and combining the acceptor [58,59].

We further utilized AFM to analyze the surface morphology and phase of binary and ternary films prepared under optimal device conditions. As shown in Fig. 5a, the PFD:Y6 blend film has a uniform surface morphology with a root-mean-square (RMS) roughness of 1.44 nm, and a fine fiber texture (Fig. 5c). After adding BTMe-4Cl, the RMS value of the blend film is enhanced to 1.96 nm, with a more robust fiber structure, which indicates that the third component BTMe-4Cl can improve the molecular aggregation of the blended film [53], in agreement with their compatibility from contact angle measurement. The consequence is conducive to charge transport.

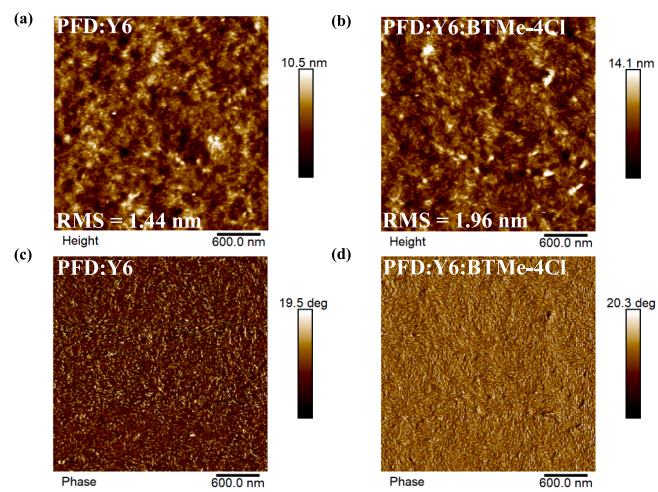


Fig. 5. AFM height and phase images of PFD:Y6 (a and c) and PFD:Y6:BTMe-4Cl blend films (b and d).

4. Conclusions

A new polymer donor, namely PFD, was designed and synthesized by combining the advantages of BDF donor units in the main chain and chlorine-substituted side chains. The prepared polymer donor PFD has a complementary absorption spectrum and matched energy levels with the star acceptor Y6. The PFD:Y6 based devices exhibited high exciton dissociation and charge collection probability, as well as low charge recombination, so a decent PCE of 16.95 % was obtained. In addition, the PFD-based devices reveal excellent long-term storage stability, which is the premise of the practical applications. The ternary stratagem of applying a small molecule acceptor BTMe-4Cl resulted in an improved nanophase separation morphology, which increased both V_{oc} and J_{sc} in the ternary device, and thus achieved a higher PCE of 17.54 %. The results prove that the performances of BDF polymer-based devices are comparable to that of BDT polymer-based ones through the rational molecule structure design.

CRediT authorship contribution statement

Jin Luo: Writing – original draft, Investigation. **Lei Wang:** Formal analysis, Data curation. **Linglong Ye:** Investigation, Funding acquisition. **Songting Tan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113766.

Data availability

No data was used for the research described in the article.

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